

Linear Algebra and Ordinary Differential Equations

MATH-121

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Solve $-2x + 3y = -9$ and $x + y = 2$.

Elimination Method

$$-2x + 3y = -9 \quad (1)$$

$$x + y = 2 \quad (2)$$

Multiplying (2) by two and adding in (1), we get

$$y = -1,$$

and

$$x = 3$$

Thus $(3, -1)$ is the solution of equation.

Substitution Method

$$-2x + 3y = -9 \quad (3)$$

$$x + y = 2 \quad (4)$$

Substituting $x = 2 - y$ from (4) into (3), we get

$$2(2 - y) + 3y = -9,$$

which gives $y = -1$
and $x = 3$.

Thus $(3, -1)$ is the solution of equation.

Matrix Method

$$AX = B \text{ or } X = A^{-1}B$$

where

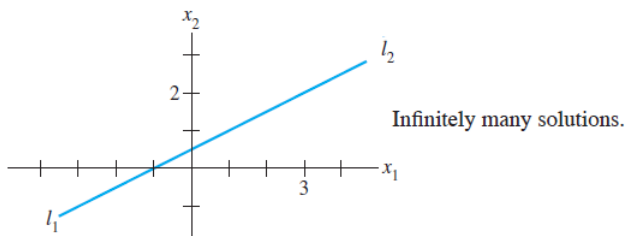
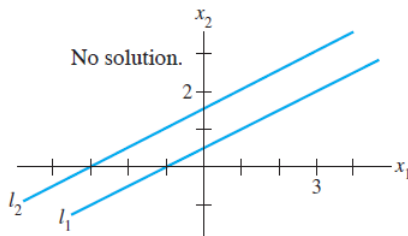
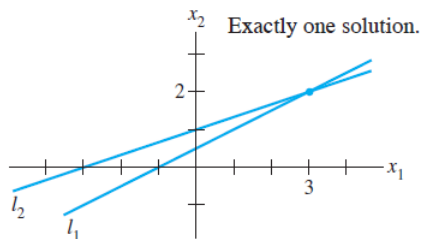
$$A = \begin{bmatrix} -2 & 3 \\ 1 & 1 \end{bmatrix},$$

X and B are given as

$$\begin{bmatrix} x \\ y \end{bmatrix} \& \begin{bmatrix} -9 \\ 2 \end{bmatrix},$$

respectively.

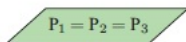
Solution of a system of Two Straight Lines



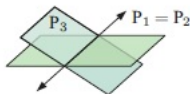
Solution of a system of Planes

- Three planes in space could have any of the following eight arrangements:

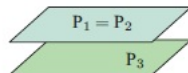
(1) all coincident



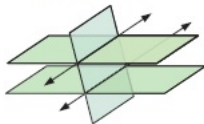
(2) two coincident and one intersecting



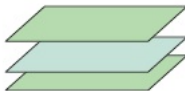
(3) two coincident and one parallel



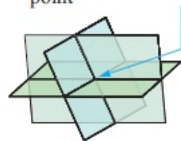
(4) two parallel and one intersecting



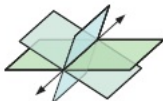
(5) all three parallel



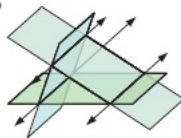
(6) all meet at the one point



(7) all meet in a common line



(8) the line of intersection of any two is parallel to the third plane.



Solution of Linear System of Equations X

$$\begin{array}{rcl} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ -4x_1 + 5x_2 + 9x_3 = -9 \end{array} \Rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ -4 & 5 & 9 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ -9 \end{bmatrix}$$

3 variables, 3 equations

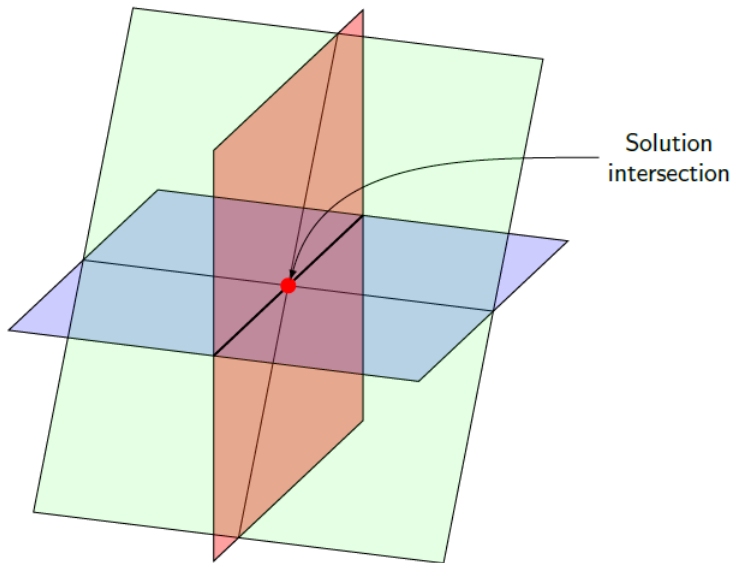
The solution of the system is

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 29 \\ 16 \\ 3 \end{bmatrix},$$

may be written as

$$X = [x_1 \quad x_2 \quad x_3]^T = [29 \quad 16 \quad 3]^T$$

Solution of a system of Planes



Linear system of m equations with n variables

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & + & \vdots & + & \ddots & + & \vdots & = & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

Note that solution of this system would be denoted as

$$X = [x_1, x_2, \cdots, x_{n-1}, x_n]^T$$

Linear System, Coefficient Matrix, Augmented Matrix

A linear system of m equations in n unknowns $x_1, x_2, \dots, x_n$ is a set of equations of the form

$$a_{11}x_1 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + \dots + a_{2n}x_n = b_2$$

$$\dots\dots\dots$$

$$a_{m1}x_1 + \dots + a_{mn}x_n = b_m.$$

Linear System, Coefficient Matrix, Augmented Matrix

A linear system of m equations in n unknowns $x_1, x_2, \dots, x_n$ is a set of equations of the form

[illegible]

$\mathbf{Ax} = \mathbf{b}$ where the coefficient matrix $\mathbf{A} = [a_{jk}]_{m \times n}$
 $\mathbf{x}$ and $\mathbf{b}$ are column vectors.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ \vdots \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ \vdots \\ b_m \end{bmatrix}_{m \times 1}$$

An Augmented Matrix determine the system completely.

$$\mathbf{A}_b = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ . & \cdots & . & . \\ . & \cdots & . & . \\ a_{m1} & \cdots & a_{mn} & b_m \end{array} \right]$$

Row Operations and Elimination Methods are Similar

$$\begin{array}{rcl} x_1 - 2x_2 + x_3 & = & 0 \\ 2x_2 - 8x_3 & = & 8 \\ -4x_1 + 5x_2 + 9x_3 & = & -9 \end{array} \quad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{array} \right]$$

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$$\begin{array}{rcl} 4 \cdot [\text{equation 1}]: & 4x_1 - 8x_2 + 4x_3 & = 0 \\ + [\text{equation 3}]: & -4x_1 + 5x_2 + 9x_3 & = -9 \\ \hline [\text{new equation 3}]: & -3x_2 + 13x_3 & = -9 \end{array}$$

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The result of this calculation is written in place of the original third equation:

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Now, multiply equation 2 by 1/2 in order to obtain 1 as the coefficient for x_2 . (This calculation will simplify the arithmetic in the next step.)

Row Operations and Elimination Methods are Similar

$$x_1 - 2x_2 + x_3 = 0$$

$$x_2 - 4x_3 = 4$$

$$-3x_2 + 13x_3 = -9$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{bmatrix}$$

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$$x_1 - 2x_2 + x_3 = 0$$

$$x_2 - 4x_3 = 4$$

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$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{bmatrix}$$

$$3 \cdot [\text{equation 2}]:$$

$$3x_2 - 12x_3 = 12$$

$$+ [\text{equation 3}]:$$

$$-3x_2 + 13x_3 = -9$$

$$\hline [\text{new equation 3}]:$$

$$\hline x_3 = 3$$

Row Operations and Elimination Methods are Similar

$$\begin{array}{rcl} x_1 - 2x_2 + x_3 & = & 0 \\ x_2 - 4x_3 & = & 4 \\ -3x_2 + 13x_3 & = & -9 \end{array} \quad \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{bmatrix}$$

$$\begin{array}{rcl} 3 \cdot [\text{equation 2}]: & 3x_2 - 12x_3 = 12 \\ + [\text{equation 3}]: & -3x_2 + 13x_3 = -9 \\ \hline [\text{new equation 3}]: & x_3 = 3 \end{array}$$

The new system has a *triangular* form:

$$\begin{array}{rcl} x_1 - 2x_2 + x_3 & = & 0 \\ x_2 - 4x_3 & = & 4 \\ x_3 & = & 3 \end{array} \quad \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

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The new system has a *triangular* form:

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$$\begin{array}{rcl} 4x_3 & = & 12 \\ x_2 - 4x_3 & = & 4 \\ \hline x_2 & = & 16 \end{array} \quad \begin{array}{rcl} -x_3 & = & -3 \\ x_1 - 2x_2 + x_3 & = & 0 \\ \hline x_1 - 2x_2 & = & -3 \end{array}$$

Row Operations and Elimination Methods are Similar

It is convenient to combine the results of these operations:

$$\begin{array}{rcl} x_1 - 2x_2 & = & -3 \\ x_2 & = & 16 \\ x_3 & = & 3 \end{array} \quad \begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

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It is convenient to combine the results of these operations:

$$\begin{array}{rcl} x_1 - 2x_2 & = & -3 \\ x_2 & = & 16 \\ x_3 & = & 3 \end{array} \quad \begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\begin{cases} x_1 & = & 29 \\ x_2 & = & 16 \\ x_3 & = & 3 \end{cases} \quad \begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Row Operations and Elimination Methods are Similar

It is convenient to combine the results of these operations:

$$\begin{array}{rcl} x_1 - 2x_2 & = & -3 \\ x_2 & = & 16 \\ x_3 & = & 3 \end{array} \quad \begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\begin{cases} x_1 & = & 29 \\ x_2 & = & 16 \\ x_3 & = & 3 \end{cases} \quad \begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

To verify that $(29, 16, 3)$ is a solution, substitute these values into the original system, and compute:

$$(29) - 2(16) + (3) = 29 - 32 + 3 = 0$$

$$2(16) - 8(3) = 32 - 24 = 8$$

$$-4(29) + 5(16) + 9(3) = -116 + 80 + 27 = -9$$

The results agree with the right side of the original system, so $(29, 16, 3)$ is a solution of the system.

ELEMENTARY ROW OPERATIONS

1. (Replacement) Replace one row by the sum of itself and a multiple of another row.
2. (Interchange) Interchange two rows.
3. (Scaling) Multiply all entries in a row by a nonzero constant.

Output of Row Operations

We apply Row operations to convert the matrix into **Echelon** or **Reduce Echelon** Form.

We apply Row operations to convert the matrix into **Echelon or Reduce Echelon** Form.

What is an **Echelon Form**?

Row Operations

Original Matrix $\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$

Echelon Form $\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$

Reduced Echelon Form $\begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$

Row Operations

$$\begin{array}{rrcrcl} 2x_1 & + & 4x_2 & + & 6x_3 & = & 18 \\ 4x_1 & + & 5x_2 & + & 6x_3 & = & 24 \\ 3x_1 & + & x_2 & - & 2x_3 & = & 4 \end{array} \quad \left(\begin{array}{ccc|c} 2 & 4 & 6 & 18 \\ 4 & 5 & 6 & 24 \\ 3 & 1 & -2 & 4 \end{array} \right)$$

This is **Row Echelon Form**

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right)$$

This is **Reduced Row Echelon Form**

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array} \right)$$

Row Operations

$$\begin{array}{l} x + 2y + 3z = 9 \\ 2x - 2z = -2 \\ 3x + 2y + z = 7 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 2 & 0 & 2 & -2 \\ 3 & 2 & 1 & 7 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{array}{l} 3x - 2y + z = 4 \\ x + 3y - z = -3 \\ 4x - 10y + 4z = 10 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 3 & -1 & -3 \\ 0 & -11 & 4 & 13 \\ 0 & 0 & 0 & -4 \end{array} \right]$$

Leading entry or leading coefficient

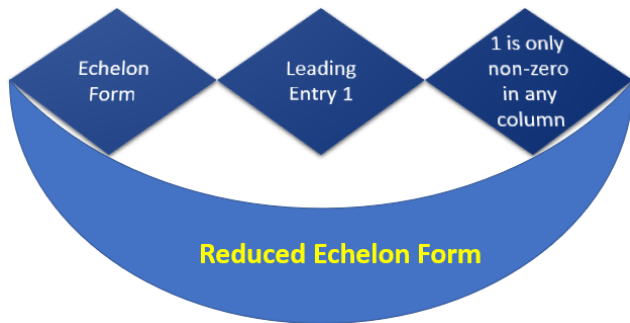
- The first non-zero element in each row, called the leading entry
- **leftmost nonzero entry**

A matrix is in row echelon form if it meets the following requirements:

- 1 Leading entry in **any row** is always right to leading entry of a row above **it**.
- 2 Rows consisting of all zeros are at the bottom of the matrix.

Reduced Echelon Form

- 1 The first non-zero number in the first row (the leading entry) is the number 1.
- 2 The second row also starts with the number 1, which is further to the right than the leading entry in the first row. For every subsequent row, the number 1 must be further to the right.
- 3 The leading entry in each row must be the only non-zero number in its column.
- 4 Any non-zero rows are placed at the bottom of the matrix.



leading entry R1

leading entry R2

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

leading entry R1

leading entry R2

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

leading entry R1

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

leading entry R2

$$\begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

$\blacksquare$ is leading entry $*$ any value including zero

In each column $\blacksquare$ is to the right of $\blacksquare$ in above row

leading entry R1

leading entry R2

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

$\blacksquare$ is leading entry $*$ any value including zero

In each column $\blacksquare$ is to the right of $\blacksquare$ in above row

The following matrices are in reduced echelon form because the leading entries are 1's, and there are 0's below *and* above each leading 1.

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

The original system of m equations in n unknowns has augmented matrix $[A|b]$. This is to be row reduced to matrix $[R|f]$. The two systems $Ax = b$ and $Rx = f$ are equivalent: if either one has a solution, so does the other, and the solutions are identical.

$$\left[\begin{array}{cccc|c} r_{11} & r_{12} & \cdots & r_{1n} & f_1 \\ & r_{22} & \cdots & r_{2n} & f_2 \\ & & \ddots & \vdots & \vdots \\ & & & r_{pp} & f_p \\ & & & \cdots & f_{p+1} \\ & & & r_{pn} & \vdots \\ & & & & f_m \end{array} \right]$$

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Echelon Form makes system easy to solve. But how to reduce a Matrix in Echelon Form?

The original system of m equations in n unknowns has augmented matrix $[A|b]$. This is to be row reduced to matrix $[R|f]$. The two systems $Ax = b$ and $Rx = f$ are equivalent: if either one has a solution, so does the other, and the solutions are identical.

$$\left[\begin{array}{cccc|c} r_{11} & r_{12} & \cdots & r_{1n} & f_1 \\ & r_{22} & \cdots & r_{2n} & f_2 \\ & & \ddots & \vdots & \vdots \\ & & & r_{pp} & f_p \\ & & & \cdots & f_{p+1} \\ & & & & \vdots \\ & & & & f_m \end{array} \right]$$

Echelon Form makes system easy to solve. But how to reduce a Matrix in Echelon Form?

The Row Reduction Algorithm

The Row Reduction Algorithm

STEP 1

Begin with the leftmost nonzero column. This is a pivot column. The pivot position is at the top.

STEP 2

Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the pivot position.

STEP 3

Use row replacement operations to create zeros in all positions below the pivot.

STEP 4

Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Apply steps 1–3 to the submatrix that remains. Repeat the process until there are no more nonzero rows to modify.

STEP 5

Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot. If a pivot is not 1, make it 1 by a scaling operation.

EXAMPLE Apply elementary row operations to transform the following matrix first into echelon form and then into reduced echelon form:

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

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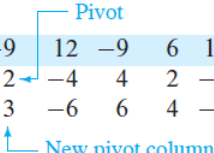
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STEP 3

Use row replacement operations to create zeros in all positions below the pivot.

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

STEP 4 Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Apply steps 1–3 to the submatrix that remains. Repeat the process until there are no more nonzero rows to modify.


$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

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Pivot

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

New pivot column

R3 ← (3/2)R2

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

STEP 4 Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Apply steps 1–3 to the submatrix that remains. Repeat the process until there are no more nonzero rows to modify.

Pivot

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

New pivot column

$R3 - (3/2)R2$

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Steps 1–3 require no work for this submatrix, and we have reached an echelon form of the full matrix. If we want the reduced echelon form, we perform one more step.

STEP 5 Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot. If a pivot is not 1, make it 1 by a scaling operation.

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 0 & -9 \\ 0 & 2 & -4 & 4 & 0 & -14 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

← Row 1 + $(-6) \cdot$ row 3

← Row 2 + $(-2) \cdot$ row 3

The next pivot is in row 2. Scale this row, dividing by the pivot.

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 0 & -9 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad \leftarrow \text{Row scaled by } \frac{1}{2}$$

Create a zero in column 2 by adding 9 times row 2 to row 1.

$$\begin{bmatrix} 3 & 0 & -6 & 9 & 0 & -72 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad \leftarrow \text{Row 1} + (9) \cdot \text{row 2}$$

Finally, scale row 1, dividing by the pivot, 3.

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad \leftarrow \text{Row scaled by } \frac{1}{3}$$

This is the reduced echelon form of the original matrix.

Using the elementary row operations on the augmented matrix

$$M = \begin{bmatrix} 1 & 2 & 0 & -1 & 1 & -10 \\ 1 & 3 & 1 & 1 & -1 & -9 \\ 2 & 5 & 1 & 0 & 0 & -19 \\ 3 & 6 & 0 & 0 & -6 & -27 \\ 1 & 3 & 3 & 5 & -5 & -7 \end{bmatrix}$$

Using the elementary row operations on the augmented matrix

$$M = \begin{bmatrix} 1 & 2 & 0 & -1 & 1 & -10 \\ 1 & 3 & 1 & 1 & -1 & -9 \\ 2 & 5 & 1 & 0 & 0 & -19 \\ 3 & 6 & 0 & 0 & -6 & -27 \\ 1 & 3 & 3 & 5 & -5 & -7 \end{bmatrix}$$

$$\begin{pmatrix} 1 & 0 & -2 & 0 & -10 & -7 \\ 0 & 1 & 1 & 0 & 4 & -1 \\ 0 & 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Write Echelon Form, using the elementary row operations on the augmented matrix

$$A_b = \begin{bmatrix} 1 & 2 & 0 & -1 & 1 & -10 \\ 1 & 3 & 1 & 1 & -1 & -9 \\ 2 & 5 & 1 & 0 & 0 & -19 \\ 3 & 6 & 0 & 0 & -6 & -27 \\ 1 & 3 & 3 & 5 & -5 & -7 \end{bmatrix}.$$

$$\begin{pmatrix} 1 & 0 & -2 & 0 & -10 & -7 \\ 0 & 1 & 1 & 0 & 4 & -1 \\ 0 & 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\left[\begin{array}{cccc|c} r_{11} & r_{12} & \cdots & r_{1n} & f_1 \\ & r_{22} & \cdots & r_{2n} & f_2 \\ & & \ddots & \vdots & \vdots \\ & & & r_{pp} & f_p \\ & & & \cdots & f_{p+1} \\ & & & r_{pn} & \vdots \\ & & & & f_m \end{array} \right]$$

The two systems $Ax = b$ and $Rx = f$ are equivalent: The row echelon form of the augmented matrix will be

$$\left[\begin{array}{cccc|c} r_{11} & r_{12} & \cdots & r_{1n} & f_1 \\ & r_{22} & \cdots & r_{2n} & f_2 \\ & & \ddots & \vdots & \vdots \\ & & & r_{pn} & f_p \\ & & & & f_{p+1} \\ & & & & \vdots \\ & & & & f_m \end{array} \right]$$

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The two systems $Ax = b$ and $Rx = f$ are equivalent: The row echelon form of the augmented matrix will be

$$\left[\begin{array}{cccc|c} r_{11} & r_{12} & \cdots & r_{1n} & f_1 \\ & r_{22} & \cdots & r_{2n} & f_2 \\ & & \ddots & \vdots & \vdots \\ & & & r_{pn} & f_p \\ & & & & f_{p+1} \\ & & & & \vdots \\ & & & & f_m \end{array} \right]$$

Rank of A_b = Number of non-zero rows of A_b in Echelon/Reduced Echelon form

Rank of A_b is denoted by $r(A_b)$.

$$\begin{cases} x + y + z = 5 \\ 2x + 3y + 5z = 8 \\ 4x + 5z = 2 \end{cases}$$

The augmented matrix of the system is the following.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 2 & 3 & 5 & 8 \\ 4 & 0 & 5 & 2 \end{array} \right]$$

Echelon Form $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 13 & -26 \end{array} \right]$

Reduced Echelon Form $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \end{array} \right]$

Gauss Elimination Method and Gauss-Jordan Elimination Method

Gauss Elimination Method = Echelon Form

+

Backward Substitution

Gauss-Jordan Elimination Method = **Reduced** Echelon Form

+

Backward Substitution

USING ROW REDUCTION TO SOLVE A LINEAR SYSTEM

1. Write the augmented matrix of the system.
2. Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent. If there is no solution, stop; otherwise, go to the next step.
3. Continue row reduction to obtain the reduced echelon form.
4. Write the system of equations corresponding to the matrix obtained in step 3.
5. Rewrite each nonzero equation from step 4 so that its one basic variable is expressed in terms of any free variables appearing in the equation.

$$\begin{bmatrix} 1 & 2 & 3 & 9 \\ 2 & 0 & -2 & -2 \\ 3 & 2 & 1 & 7 \end{bmatrix}$$

$$R_2 = R_2 - 2R_1$$

$$R_3 = R_3 - 3R_1$$

$$R_2 = -\frac{R_2}{4}$$

$$R_1 = R_1 - 2R_2$$

$$R_3 = R_3 + 4R_2$$

$$\begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Determining whether $Ax = b$ has solutions:

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Solve the following systems

1

$$x + 2y + 3z = 9$$

$$2x - 2z = -2$$

$$3x + 2y + z = 7$$

Determining whether $Ax = b$ has solutions:

Solve the following systems

1

$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad A_b = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 2 & 0 & -2 & -2 \\ 3 & 2 & 1 & 7 \end{bmatrix}$$

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$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad A_b = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 2 & 0 & -2 & -2 \\ 3 & 2 & 1 & 7 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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2

$$3x - 2y + z = -6$$

$$4x - 3y + 3z = 7$$

$$2x + y - z = -9$$

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2

$$3x - 2y + z = -6$$

$$4x - 3y + 3z = 7$$

$$2x + y - z = -9$$

$$A_b = \begin{bmatrix} 3 & -2 & 1 & -6 \\ 4 & -3 & 3 & 7 \\ 2 & 1 & -1 & -9 \end{bmatrix}$$

Determining whether $Ax = b$ has solutions:

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$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad A_b = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 2 & 0 & -2 & -2 \\ 3 & 2 & 1 & 7 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2

$$\begin{array}{rcl} 3x - 2y + z & = & -6 \\ 4x - 3y + 3z & = & 7 \\ 2x + y - z & = & -9 \end{array} \quad A_b = \begin{bmatrix} 3 & -2 & 1 & -6 \\ 4 & -3 & 3 & 7 \\ 2 & 1 & -1 & -9 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \end{bmatrix}$$

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$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad A_b = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 2 & 0 & -2 & -2 \\ 3 & 2 & 1 & 7 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2

$$\begin{array}{rcl} 3x - 2y + z & = & -6 \\ 4x - 3y + 3z & = & 7 \\ 2x + y - z & = & -9 \end{array} \quad A_b = \begin{bmatrix} 3 & -2 & 1 & -6 \\ 4 & -3 & 3 & 7 \\ 2 & 1 & -1 & -9 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \end{bmatrix}$$

3

$$\begin{array}{rcl} 3x - 2y + z & = & 4 \\ x + 3y - z & = & -3 \\ 4x - 10y + 4z & = & 10 \end{array}$$

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Solve the following systems

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$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad A_b = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 2 & 0 & -2 & -2 \\ 3 & 2 & 1 & 7 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2

$$\begin{array}{rcl} 3x - 2y + z & = & -6 \\ 4x - 3y + 3z & = & 7 \\ 2x + y - z & = & -9 \end{array} \quad A_b = \begin{bmatrix} 3 & -2 & 1 & -6 \\ 4 & -3 & 3 & 7 \\ 2 & 1 & -1 & -9 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \end{bmatrix}$$

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$$\begin{array}{rcl} 3x - 2y + z & = & 4 \\ x + 3y - z & = & -3 \\ 4x - 10y + 4z & = & 10 \end{array} \quad \begin{bmatrix} 3 & -2 & 1 & 4 \\ 1 & 3 & 1 & -3 \\ 4 & -10 & 4 & 10 \end{bmatrix}$$

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Solve the following systems

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$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad A_b = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 2 & 0 & -2 & -2 \\ 3 & 2 & 1 & 7 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2

$$\begin{array}{rcl} 3x - 2y + z & = & -6 \\ 4x - 3y + 3z & = & 7 \\ 2x + y - z & = & -9 \end{array} \quad A_b = \begin{bmatrix} 3 & -2 & 1 & -6 \\ 4 & -3 & 3 & 7 \\ 2 & 1 & -1 & -9 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \end{bmatrix}$$

3

$$\begin{array}{rcl} 3x - 2y + z & = & 4 \\ x + 3y - z & = & -3 \\ 4x - 10y + 4z & = & 10 \end{array} \quad \begin{bmatrix} 3 & -2 & 1 & 4 \\ 1 & 3 & 1 & -3 \\ 4 & -10 & 4 & 10 \end{bmatrix} \quad \begin{bmatrix} 1 & 3 & -1 & -3 \\ 0 & -11 & 4 & 13 \\ 0 & 0 & 0 & -4 \end{bmatrix}$$

$r(A_b) = r(A) \implies$ Existence of solution

1

$$x + 2y + 3z = 9$$

$$2x - 2z = -2$$

$$3x + 2y + z = 7$$

$r(A_b) = r(A) \implies$ Existence of solution

①

$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad r(A_b) = 2 = r(A)$$

$r(A_b) = r(A) \implies$ Existence of solution

1

$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad r(A_b) = 2 = r(A)$$

2

$$\begin{array}{rcl} 3x - 2y + z & = & -6 \\ 4x - 3y + 3z & = & 7 \\ 2x + y - z & = & -9 \end{array}$$

$r(A_b) = r(A) \implies$ Existence of solution

1

$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad r(A_b) = 2 = r(A)$$

2

$$\begin{array}{rcl} 3x - 2y + z & = & -6 \\ 4x - 3y + 3z & = & 7 \\ 2x + y - z & = & -9 \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \end{bmatrix} \quad r(A_b) = 3 = r(A)$$

$r(A_b) = r(A) \implies$ Existence of solution

1

$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad r(A_b) = 2 = r(A)$$

2

$$\begin{array}{rcl} 3x - 2y + z & = & -6 \\ 4x - 3y + 3z & = & 7 \\ 2x + y - z & = & -9 \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \end{bmatrix} \quad r(A_b) = 3 = r(A)$$

3

$$\begin{array}{rcl} 3x - 2y + z & = & 4 \\ x + 3y - z & = & -3 \\ 4x - 10y + 4z & = & 10 \end{array}$$

$r(A_b) = r(A) \implies$ Existence of solution

1

$$\begin{array}{rcl} x + 2y + 3z & = & 9 \\ 2x - 2z & = & -2 \\ 3x + 2y + z & = & 7 \end{array} \quad \begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad r(A_b) = 2 = r(A)$$

2

$$\begin{array}{rcl} 3x - 2y + z & = & -6 \\ 4x - 3y + 3z & = & 7 \\ 2x + y - z & = & -9 \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \end{bmatrix} \quad r(A_b) = 3 = r(A)$$

3

$$\begin{array}{rcl} 3x - 2y + z & = & 4 \\ x + 3y - z & = & -3 \\ 4x - 10y + 4z & = & 10 \end{array} \quad \begin{bmatrix} 1 & 3 & -1 & -3 \\ 0 & -11 & 4 & 13 \\ 0 & 0 & 0 & -4 \end{bmatrix} \quad r(A_b) = 3, \quad r(A) = 2$$

Determining whether $Ax = b$ has solutions:

Solve the following systems Comparing rank of Coefficient Matrix A and rank of Augmented Matrix A_b for these systems we obtain

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Solve the following systems Comparing rank of Coefficient Matrix A and rank of Augmented Matrix A_b for these systems we obtain

For First system

$$r(A) = 2 = r(A_b) \Rightarrow \text{System is consistent.}$$

For Second system

$$r(A) = 3 = r(A_b) \Rightarrow \text{System is consistent.}$$

For Third system

$$r(A) = 2 \ \& \ r(A_b) = 3 \Rightarrow \text{System is inconsistent.}$$

Determining whether $Ax = b$ has solutions:

Solve the following systems Comparing rank of Coefficient Matrix A and rank of Augmented Matrix A_b for these systems we obtain

For First system

$$r(A) = 2 = r(A_b) \Rightarrow \text{System is consistent.}$$

For Second system

$$r(A) = 3 = r(A_b) \Rightarrow \text{System is consistent.}$$

For Third system

$$r(A) = 2 \ \& \ r(A_b) = 3 \Rightarrow \text{System is inconsistent.}$$

In general,

- 1 If $r(A) = r(A_b)$ then system is consistent
- 2 If $r(A) < r(A_b)$ then system is inconsistent

ELEMENTARY ROW OPERATIONS

1. (Replacement) Replace one row by the sum of itself and a multiple of another row.
2. (Interchange) Interchange two rows.
3. (Scaling) Multiply all entries in a row by a nonzero constant.

Existence and Uniqueness Theorem

A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column—that is, if and only if an echelon form of the augmented matrix has *no* row of the form

$$[0 \ \cdots \ 0 \ b] \quad \text{with } b \text{ nonzero}$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.

Find The solution to the

$$\begin{bmatrix} 1 & 3 & 2 & 4 & -3 \\ 2 & 6 & 0 & -1 & -2 \\ 0 & 0 & 6 & 2 & -1 \\ 1 & 3 & -1 & 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -7 \\ 0 \\ 12 \\ -6 \end{bmatrix}$$

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- $R_2 \rightarrow R_2 - 2R_1$ and $R_4 \rightarrow R_4 - R_1$
- $R_3 \rightarrow 4 \times R_3$ and $R_4 \rightarrow 4 \times R_4$
- $R_3 \rightarrow R_3 + 6.R_2$ and $R_4 \rightarrow R_4 - R_2$
- $R_3 \rightarrow R_3/2$
- $R_4 \rightarrow 23R_4 + 27R_3$
- $R_4 \rightarrow R_4/454$
- $R_1 \leftarrow R_1 + 3R_4$, $R_2 \rightarrow R_2 - 4R_4$ and $R_3 \rightarrow R_3 + 10R_4$

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- $R_4 \rightarrow R_4/454$
- $R_1 \leftarrow R_1 + 3R_4$, $R_2 \rightarrow R_2 - 4R_4$ and $R_3 \rightarrow R_3 + 10R_4$
- $R_3 \rightarrow \frac{R_3}{23}$
- $R_1 \rightarrow R_1 - 4R_3$ and $R_2 \rightarrow R_2 + 9R_3$
- $R_2 \rightarrow \frac{R_2}{-4}$
- $R_1 \rightarrow R_1 - 2R_2$

Find The solution to the

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- $R_2 \rightarrow R_2 - 2R_1$ and

- $R_4 \rightarrow R_4 - R_1$

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- $R_2 \rightarrow R_2 + 9R_3$

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$$\left[\begin{array}{ccccc|c} 1 & 3 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{array} \right]$$

EXAMPLE 2 The network in Fig. 2 shows the traffic flow (in vehicles per hour) over several one-way streets in downtown Baltimore during a typical early afternoon. Determine the general flow pattern for the network.

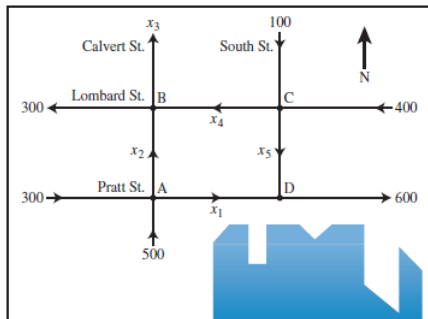
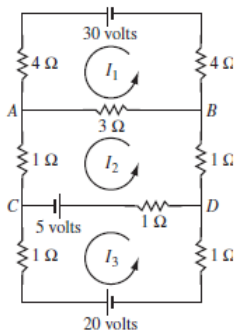


FIGURE 2 Baltimore streets.

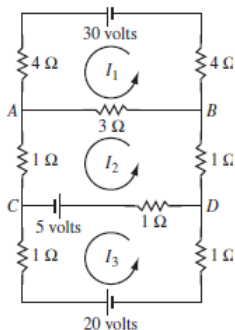
SOLUTION Write equations that describe the flow, and then find the general solution of the system. Label the street intersections (junctions) and the unknown flows in the branches, as shown in Fig. 2. At each intersection, set the flow in equal to the flow out.

Intersection	Flow in	Flow out
A	$300 + 500$	$= x_1 + x_2$
B	$x_2 + x_4$	$= 300 + x_3$
C	$100 + 400$	$= x_4 + x_5$
D	$x_1 + x_5$	$= 600$

EXAMPLE Determine the loop currents in the network in Fig.



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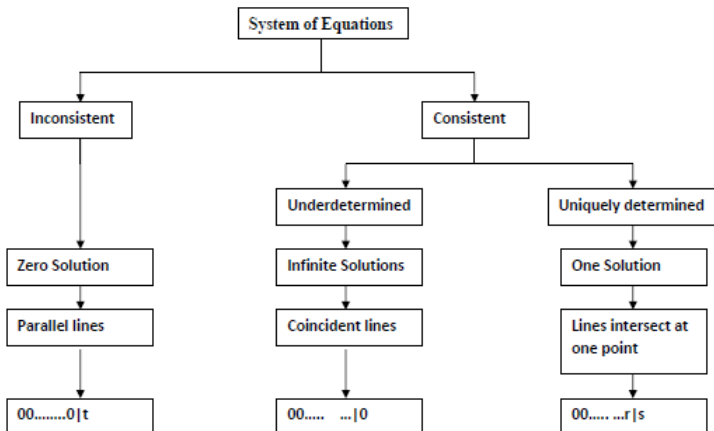


The loop currents are found by solving the system

$$11I_1 - 3I_2 = 30$$

$$-3I_1 + 6I_2 - I_3 = 5$$

$$-I_2 + 3I_3 = -25$$



QUIZ#1

Odd Rows Is the following system consistent? If yes how many solutions are there for the system?

$$\begin{aligned}x_1 - 5x_2 + 4x_3 &= -3 \\2x_1 - 7x_2 + 3x_3 &= -2 \\-2x_1 + 1x_2 + 7x_3 &= -1\end{aligned}$$

Even Rows Use row reduction algorithm to find rank then comment about existence and uniqueness of the system

$$\begin{aligned}x_2 + 5x_3 &= -4 \\x_1 + 4x_2 + 3x_3 &= -2 \\2x_1 + 7x_2 + x_3 &= -2\end{aligned}$$